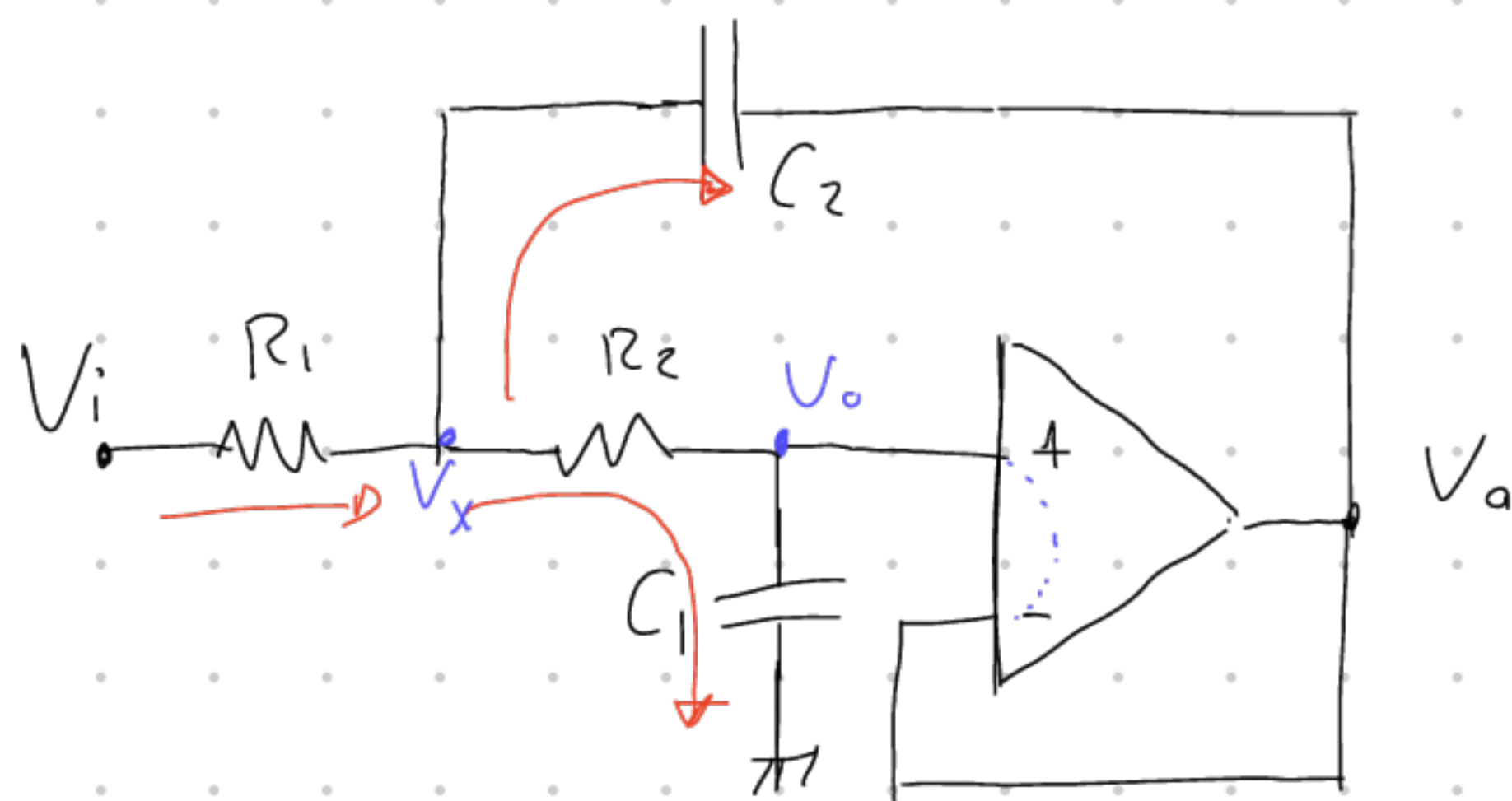


Sallen - key pasa bajos (LP)



$$\textcircled{I} (V_x - V_o)G_2 = V_o \cdot sC_1$$

$$V_x = V_o \left(\frac{sC_1}{G_2} + 1 \right)$$

$$\textcircled{II} (V_i - V_x)G_1 = V_o \cdot sC_1 + (V_x - V_o)sC_2$$

$$V_i G_1 = V_o (sC_1 - sC_2) + V_x (sC_2 + G_1)$$

$$V_i G_1 = V_o \left[sC_1 - sC_2 + \left(\frac{sC_1}{G_2} + 1 \right) (sC_2 + G_1) \right]$$

$$V_i G_1 = V_o \left[sC_1 - sC_2 + \frac{s^2 C_1 C_2 + sC_1 G_1 + sC_2 G_2 + G_1 G_2}{G_2} \right]$$

$$V_i G_1 = V_o \left(\frac{sC_1 G_2 - sC_2 G_2 + s^2 C_1 C_2 + sC_1 G_1 + sC_2 G_2 + G_1 G_2}{G_2} \right)$$

$$\frac{V_o}{V_i} = \frac{G_1 G_2}{s^2 C_1 C_2 + sC_1 (G_1 + G_2) + G_1 G_2}$$

$$T(s) = \frac{V_o}{V_i} = \frac{\frac{G_1 G_2}{C_1 C_2}}{s^2 + s \frac{1}{C_2} (G_1 + G_2) + \frac{G_1 G_2}{C_1 C_2}}$$

$$T(s) = \frac{1}{R_1 R_2 C_1 C_2} \frac{1}{s^2 + s \frac{1}{C_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$\omega_0^2 = \frac{1}{R_1 R_2 C_1 C_2} \rightarrow$$

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$\frac{\omega_0}{Q} = \frac{1}{C_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \rightarrow Q = \frac{\frac{1}{\sqrt{R_1 R_2 C_1 C_2}}}{\frac{1}{C_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)} = \frac{C_2}{\sqrt{R_1 R_2 C_1 C_2} \left(\frac{R_2 + R_1}{R_1 R_2} \right)}$$

$$Q = \frac{C_2 R_1 R_2}{\sqrt{R_1 R_2 C_1 C_2} (R_2 + R_1)} = \sqrt{\frac{C_2^2 R_2^2}{R_1 R_2 C_1 C_2}} \cdot \frac{R_1}{R_1 + R_2}$$

$$Q = \sqrt{\frac{C_2 R_2}{C_1 R_1}} \cdot \frac{R_1}{R_1 + R_2}$$